FIITJEE

ALL INDIA TEST SERIES

PART TEST – II

JEE (Main)-2021

TEST DATE: 12-12-2020

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 75 questions.
- Each subject (PCM) has 25 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only three sections: **Section-A**, **Section-B and Section-C**.

  Section-A (01 – 20, 26 – 45, 51 – 70) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 22, 46 – 47, 71 – 72) *contains 6 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

  Section-C (23 – 25, 48 – 50, 73 – 75) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

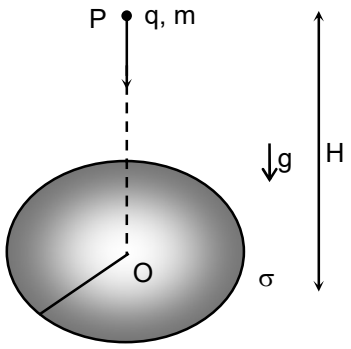
Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A particle of mass 'm' and charge 'q' is dropped, along the axis of a fixed disc from a height 'H' with zero initial velocity (as shown). The disc has a radius 'R' and a uniform charge density σ, with its axis vertical. The particle has $\left(\frac{q}{m}\right)=\left(\frac{4\varepsilon_0 g}{\sigma}\right)$. If the particle just reaches the disc, the value of 'H' is



(A) $\frac{R}{\sqrt{2}}$

(B) $\frac{R}{2}$

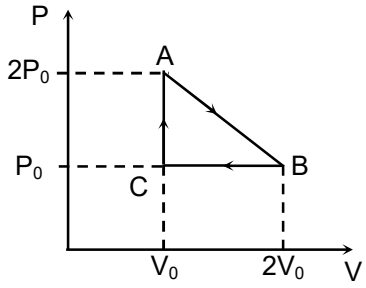
(C) $\frac{3R}{4}$

(D) $\frac{4R}{3}$

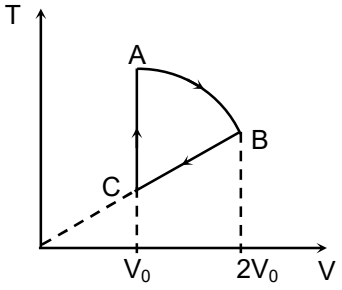
2. A rigid container, containing 3 moles of an ideal diatomic gas is heated in such a way that the temperature of the gas increases to 400 K from 300 K. Also, one mole of the diatomic gas breaks into individual free atoms. Find the net heat transferred to the gas.
(A) 950 R
(B) 450 R
(C) 800 R
(D) 400 R

3. An ice slab is melting at 273 K and atmospheric pressure. Internal energy of the ice-water system is denoted by U and work done by the ice-water system on the atmosphere is denoted by W. Then,
(A) U increases and W is positive
(B) U increases and W is negative
(C) U decreases and W is positive
(D) U decreases and W is negative

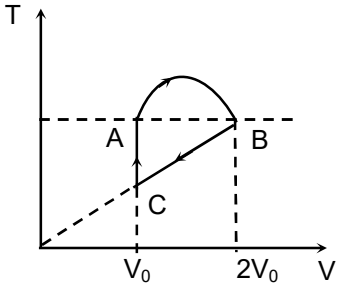
4. P-V graph of a cyclic process is as shown in the figure. Then its correct T-V curve would be



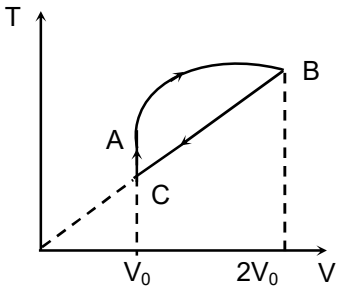
(A)



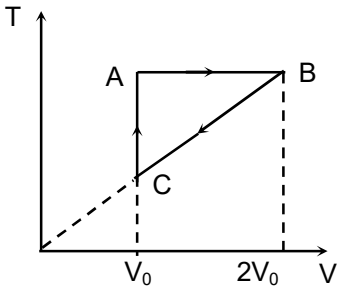
(B)



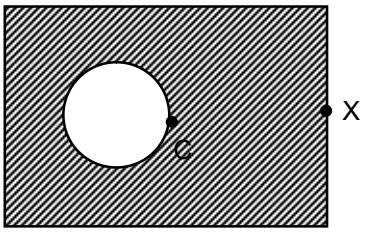
(C)



(D)



5. A uniform square plate with a circular hole lies over a smooth horizontal surface. C is the centre of the plate and X is at the edge of the plate, as shown. Now, if the temperature of the plate is increased



(A) C remains at rest and X moves right
(B) C moves left and X remains at rest
(C) Both C and X moves to the right
(D) C moves left and X moves right

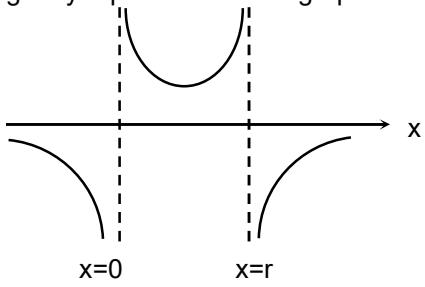
6. A small bob of mass m is suspended by means of a light metallic wire. The bob is moving in a vertical circle about a fixed point. Now, the temperature of the wire is increased and the bob continues to move in a vertical circle of increased radius. Lets take T_1 and T_2 as the values of the tension in the wire at its highest and lowest points respectively. Then, due to increase in temperature:
(A) T_1 decreases but T_2 increases
(B) T_1 increases but T_2 decreases
(C) Both T_1 and T_2 decrease
(D) Both T_1 and T_2 increase

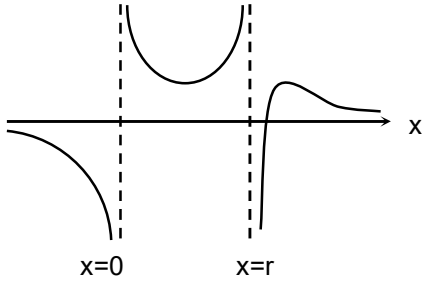
7. For a ferro-magnetic substance, the value of retentivity should be
(A) high for permanent magnets as well as core of electromagnets.
(B) low for permanent magnets as well as core of electromagnets.
(C) low for permanent magnets and high for core of electromagnets.
(D) high for permanent magnets and low for core of electromagnets.

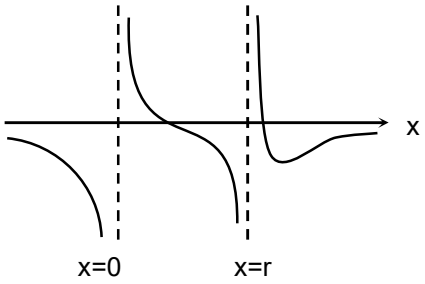
8. According to the Curie's law, the value of magnetic susceptibility of a paramagnetic substance
(A) increases with external magnetic field and decreases with absolute temperature.
(B) increases with external magnetic field as well as with absolute temperature.
(C) is not affected by the external magnetic field but decreases with absolute temperature.
(D) increases with the external magnetic field but is not affected by the absolute temperature.

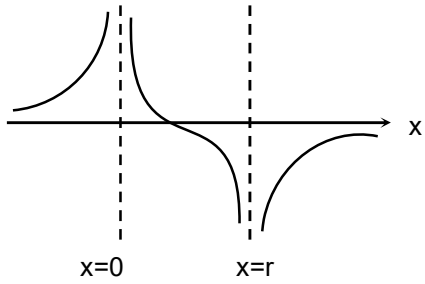
9. A hollow and a solid sphere of same material and identical outer surfaces are heated to the same temperature and are kept in identical surroundings. Then, which of the following quantities will have different values for the two of them, initially
(A) Amount of energy radiated per unit time.
(B) Amount of radiation absorbed per unit time.
(C) rate of fall of temperature (dT/dt).
(D) None of these.

10. A positive point charge 'Q' is located at origin and a negative point charge 'q' is located on x-axis at a distance 'r' from 'Q'. Magnitude of charge of 'Q' is greater than that of 'q'. Then, which of the following may represent the E-x graph:

(A)


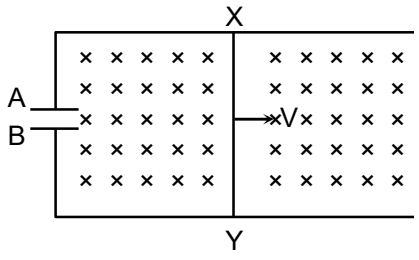
(B)


(C)


(D)


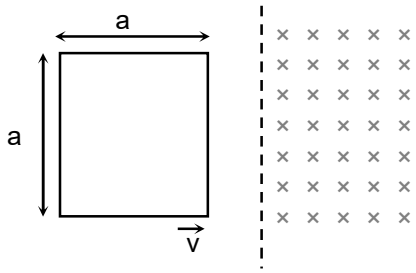
11. There exist a uniform magnetic field of 1T and a uniform electric field of 1V/m – both along $\left(-\hat{i}+\hat{j}\right)$. A charged particle of mass 1 Kg and of charge 1C is having a velocity of $\frac{1}{\sqrt{2}}\left(\hat{i}+\hat{j}\right)$ m/s and is at origin at t = 0. Then, the coordinates of the particle at time 'π' seconds will be (take $\pi^2 = 10$)
(A) (0, 5, 2) m
(B) $\left(0,\frac{5}{\sqrt{2}},-2\right)$m
(C) $\left(\frac{-5}{\sqrt{2}},\frac{5}{\sqrt{2}},2\right)$m
(D) (– 5, 5, –2) m

12. A conducting rod XY of length $\ell = 10\,m$ is moving with a uniform speed of 4 m/s in a uniform magnetic field B = 2T directed into the plane, as shown. A capacitor of capacity, C = 10μF is connected as shown in the figure. If the rod is moving in a horizontal plane, then



(A) q_A = + 800 μC & q_B = – 800 μC
(B) q_A = – 800 μC & q_B = + 800 μC
(C) $q_A = q_B = 0$
(D) Charge on the capacitor increases with time.

13. A square loop of side 'a' and resistance 'R' is moved in the region of uniform magnetic field 'B' from a region of zero magnetic field, as shown in the figure. If the loop is moved with a constant velocity 'v' through a distance 'x' inside the field, the work done by the magnetic field is



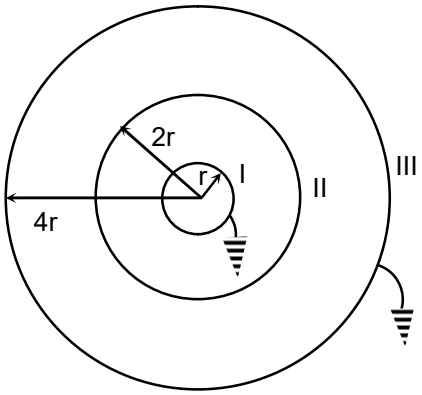
(A) $\frac{B\ell^2vx}{R}$

(B) $\frac{B\ell^2vx}{2R}$

(C) $\frac{2B\ell^2vx}{R}$

(D) Zero

14. In the shown figure, there are three concentric conducting spherical shells. Inner and outer shells are earthed and the middle shell is given a charge of 30 μC. Then



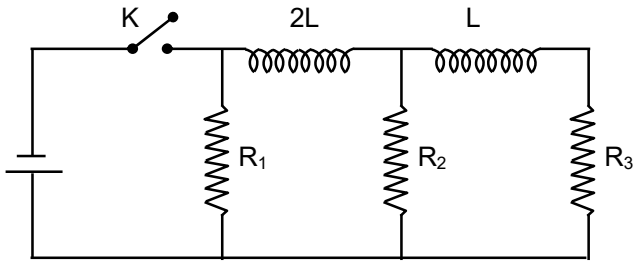
(A) Charge on the I shell is – 20 μC

(B) Charge on the III shell is – 10 μC

(C) Net charge on the system is positive

(D) Net charge on the system is zero

15. In the given circuit, the resistors are identical but the inductors have different inductance (as shown). The key is closed and after a long time the current in the third resistor was found to be I_0. If the key is now opened, then just after this moment



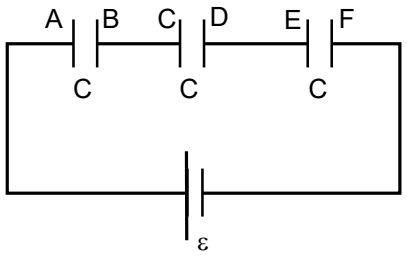
(A) Current in R_3 is $I_0/2$

(B) Current in R_2 is $3I_0/2$

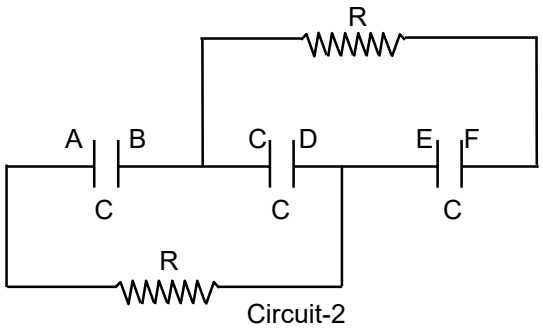
(C) Current in R_1 is $2I_0$

(D) Current through both the inductors are same.

16. Three capacitors, each of capacitance C are connected in series. The capacitors are charged by connecting a battery of emf ε, as shown in the circuit-1. Now, the battery is disconnected and two identical resistors are connected simultaneously, as shown in the circuit-2. Then, the net heat dissipated is equal to



Circuit -1



Circuit-2

(A) $\frac{8}{9}C\varepsilon^2$

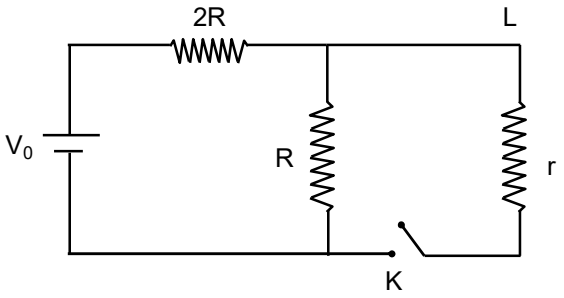
(B) $\frac{4}{27}C\varepsilon^2$

(C) $\frac{2}{27}C\varepsilon^2$

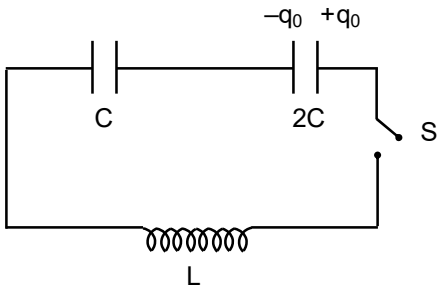
(D) $\frac{4}{9}C\varepsilon^2$

17. In the given figure, the battery is ideal. If the key is closed, the current through the '2R' resistor increases by '1A'. Then, the value of current through the 'r' resistor is:

(A) 2 A
(B) 3 A
(C) 1/2 A
(D) 1/3 A



18. The switch in the circuit below is open; the capacitor of capacitance 2C carries electric charge q_0; the capacitor of capacitance C is uncharged; and there is electric current in the inductor. Find the value of the maximum current through the switch after it is closed.



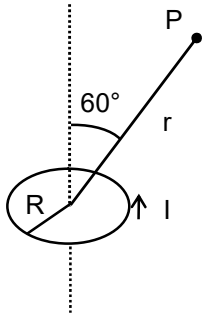
(A) $\frac{q_0}{\sqrt{6LC}}$

(B) $\frac{q_0}{\sqrt{3LC}}$

(C) $\frac{\sqrt{2}q_0}{\sqrt{3LC}}$

(D) $\frac{q_0}{\sqrt{LC}}$

19. A current carrying loop of radius R and current I is shown in the diagram. Find the magnitude of the magnetic field at point P due to this loop. (take r >> R)



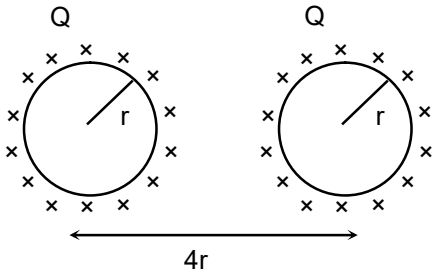
(A) $\frac{\sqrt{5}}{4}\left(\frac{\mu_0 IR^2}{r^3}\right)$

(B) $\frac{\sqrt{7}}{4}\left(\frac{\mu_0 IR^2}{r^3}\right)$

(C) $\frac{\sqrt{5}}{8}\left(\frac{\mu_0 IR^2}{r^3}\right)$

(D) $\frac{\sqrt{7}}{8}\left(\frac{\mu_0 IR^2}{r^3}\right)$

20. Two non-conducting spherical shells, each of radius r and uniformly distributed charge Q are placed at a separation of '4r' from each other as shown. An external agent brings a charge Q from infinity to the second sphere and distributes uniformly over it. Then, the work done by the external agent in this process is



(A) $\left(\frac{1}{4\pi\varepsilon_0}\right)\frac{Q^2}{4r}$

(B) $\left(\frac{1}{4\pi\varepsilon_0}\right)\frac{3Q^2}{4r}$

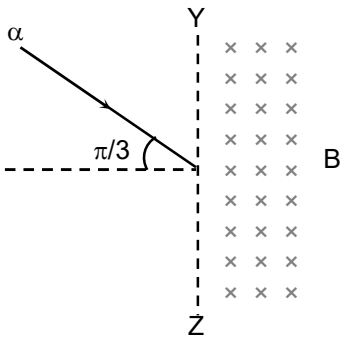
(C) $\left(\frac{1}{4\pi\varepsilon_0}\right)\frac{5Q^2}{4r}$

(D) $\left(\frac{1}{4\pi\varepsilon_0}\right)\frac{7Q^2}{4r}$

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. A particle with specific charge α, enters a magnetic field of magnitude B, existing only to the right of the boundary YZ, as shown. The direction of motion of the particle is perpendicular to the direction of B. If the time spent by the particle in the field is $K \times \frac{\pi}{3\alpha B}$, find the value of 'K'

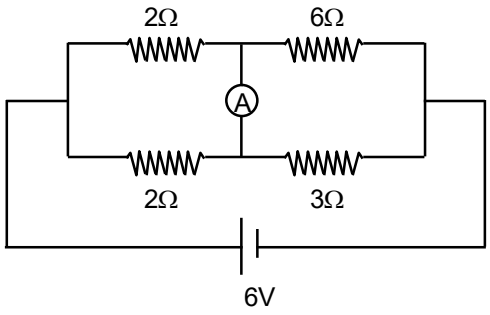


22. A gas undergoes a process such that $P \propto \frac{1}{T}$. If, the molar heat capacity for this process is 4.5 R (R universal gas constant), find the degree of freedom of the molecules of the gas.

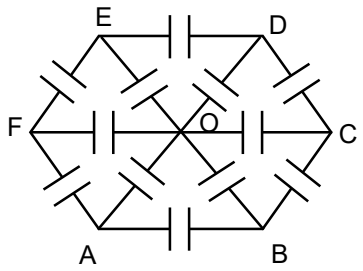
SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

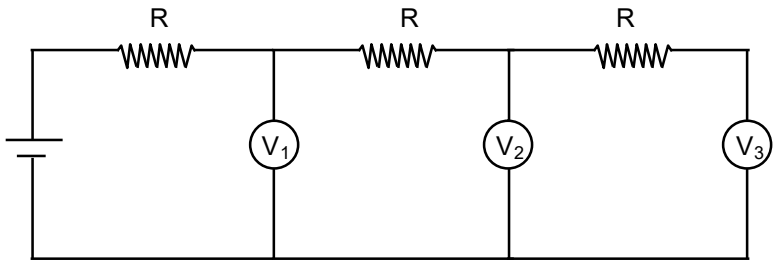
23. Determine the reading of the ideal ammeter in the following figure (in A)



24. All the capacitors in the given figure has a capacitance C_0. Find the equivalent capacitance between the points A and O in terms of C_0.



25. All the three voltmeters are identical. The first voltmeter reads, V_1 = 10 V and the third voltmeter reads, V_3 = 8 V. If the three resistors are also identical, find the reading of the second voltmeter (in volts) (Take $\sqrt{10} = 3.16$).



Chemistry

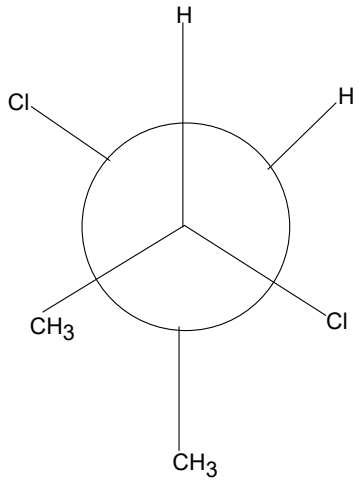
PART – II

SECTION – A
(One Options Correct Type)

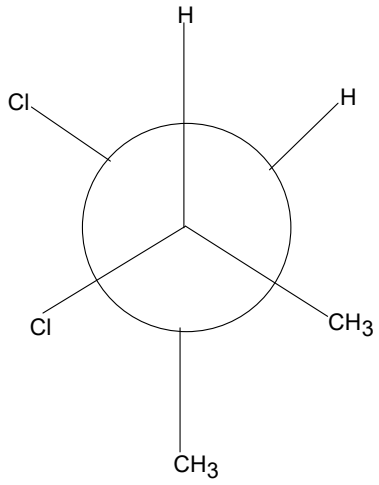
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

26. In the elimination of given vicinal dihalide with Zn/CH_3COOH, which conformation leads to formation of cis – 2 – butene?

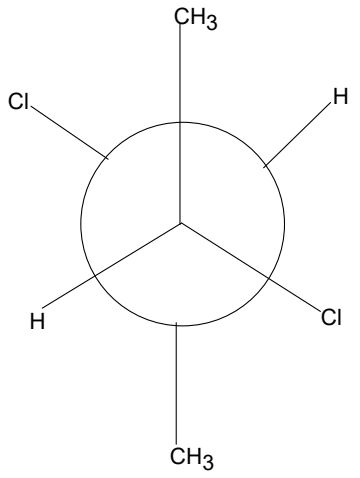
(A)



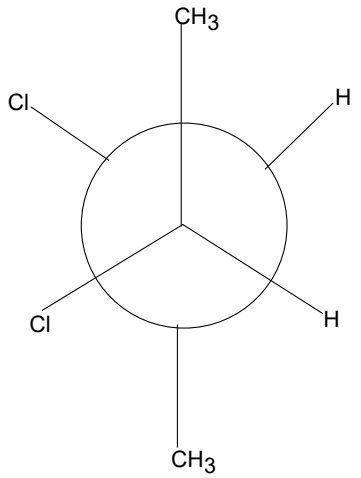
(B)



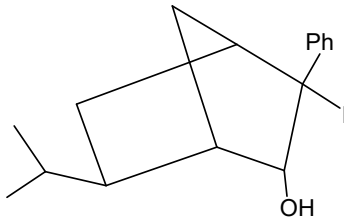
(C)



(D)

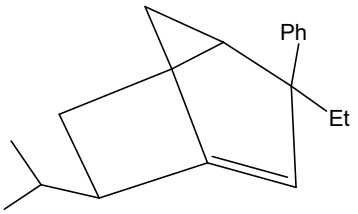


27.

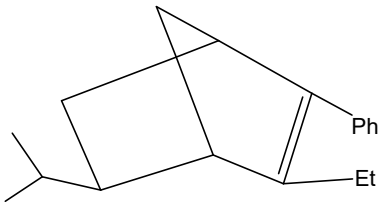


$\xrightarrow{H^+}$ The major prdoduct is

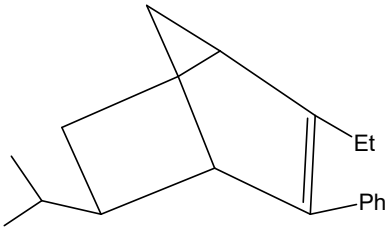
(A)



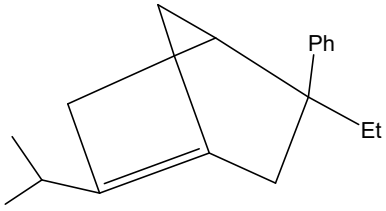
(B)



(C)

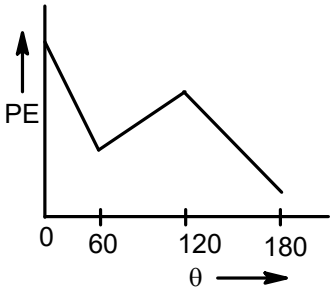


(D)

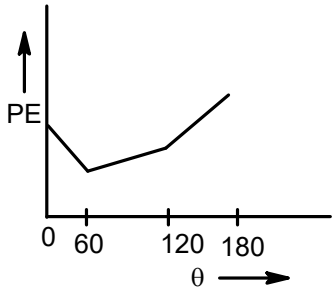


28. The energy profile diagram that corresponds to COOH – CH_2 – CH_2 – COOH for rotation around C_2 – C_3 bond in acidic medium is:

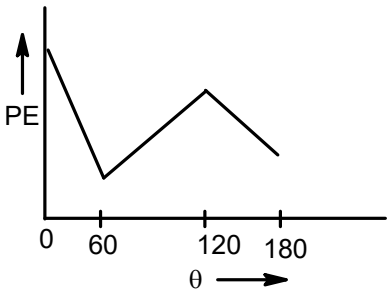
(A)



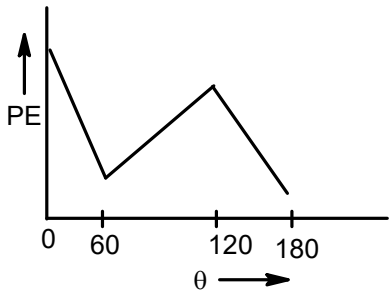
(B)



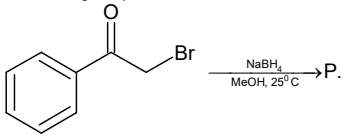
(C)



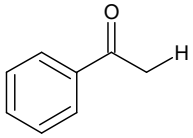
(D)



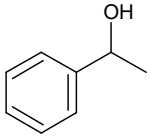
29. The major product of the following reaction is



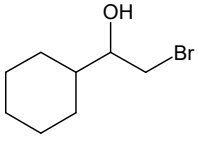
(A)



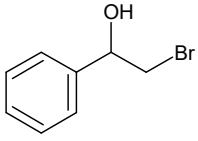
(B)



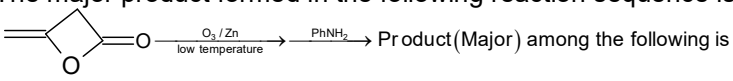
(C)

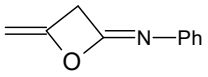


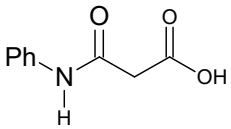
(D)



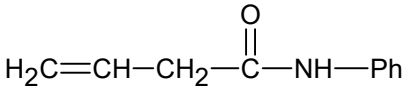
30. The major product formed in the following reaction sequence is:



(A) 

(B) 

(C) Ph – N = CH – CH_2 – COOH

(D) 

31. Which of the following tests do not confirm the presence of carbohydrate molecules?
(i) Xanthoprotic test
(ii) Molisch test
(iii) Biuret test
(iv) Barfoed test
(A) (i), (ii), (iii)
(B) (i), (iii), (iv)
(C) (i), (iii)
(D) (i), (ii), (iii), (iv)

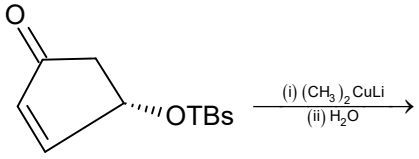
32. The type of forces present is cis – polyisopropene molecule that held them together are
(A) Weak vander Waal's
(B) H – bonding
(C) Dipole – dipole
(D) Ion – dipole

33. For the reaction:

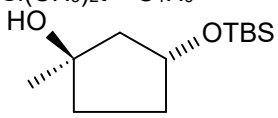
$$HX(aq) + H_2O(\ell) \rightleftharpoons H_3O^+(aq) + X^-(aq)$$

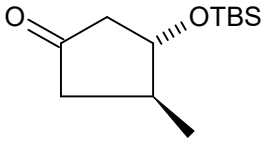
The highest value of $[X^-]$ (aq), when X^- is
(A) F^-
(B) Cl^-
(C) NO_2^-
(D) I^-

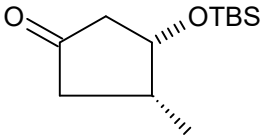
34. The major product in the following reaction is:

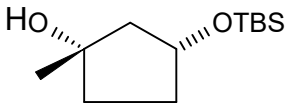


TBS = $Si(CH_3)_2t - C_4H_9$

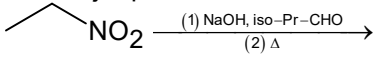
(A) 

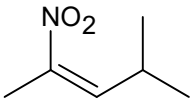
(B) 

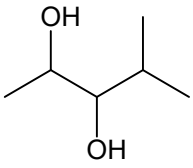
(C) 

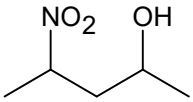
(D) 

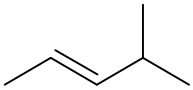
35. The major product in the following reaction is:

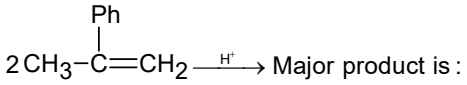


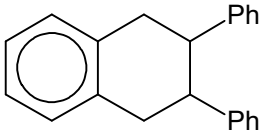
(A) 

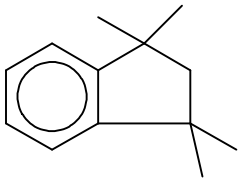
(B) 

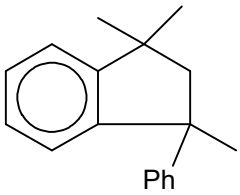
(C) 

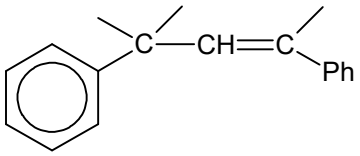
(D) 

36. 

(A) 

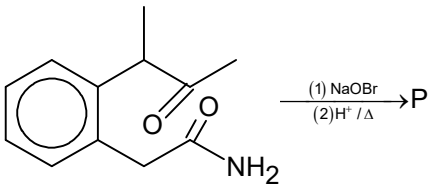
(B) 

(C) 

(D) 

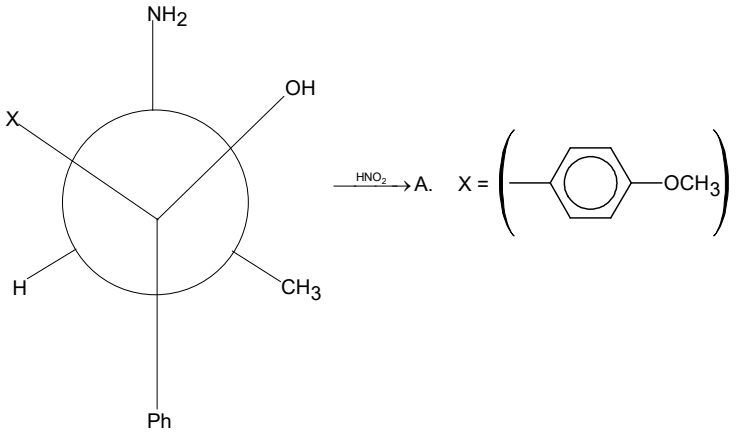
37. High blindness is caused by deficiency of
(A) Vitamin A
(B) Vitamin B_2
(C) Vitamin C
(D) Vitamin D

38. For the given reaction

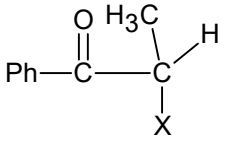


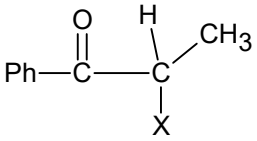
The correct statement regarding major product P is
(A) P has an acidic group (–COOH)
(B) P has an amine group ($–NH_2$)
(C) P has both (–COOH) and ($–NH_2$) group present
(D) P on hydrolysis produce a molecule with both (–COOH) and ($–NH_2$) group present

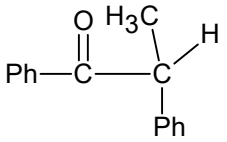
39.

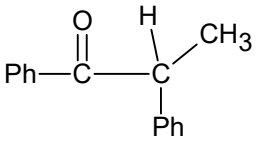


Major product A in the above reaction is

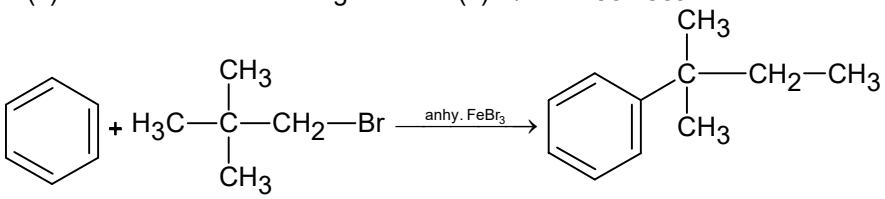
(A) 

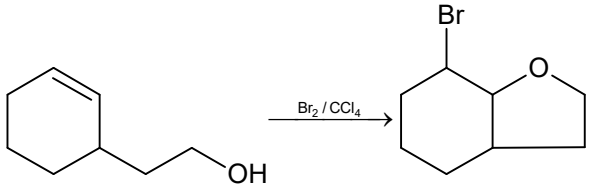
(B) 

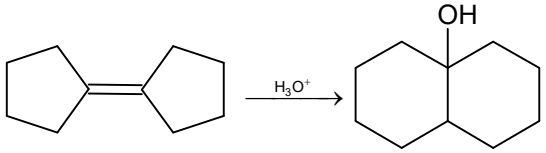
(C) 

(D) 

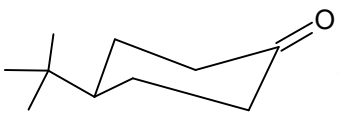
40. The product(s) of which of the following reaction (s) is/are **incorrect**?

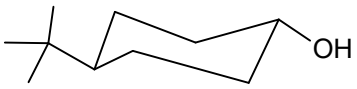
(i) 

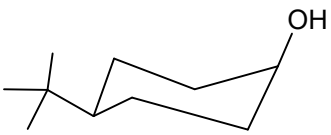
(ii) 

(iii) 

(A) (i), (iii)
(B) (ii), (iii)
(C) only (i)
(D) only (iii)

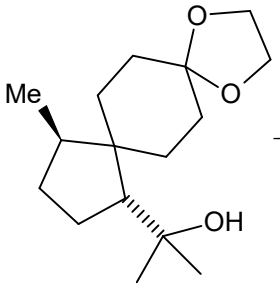
41.  $\xrightarrow[(ii)H_2O]{(i)\ LiAlH_4}$ The major product is

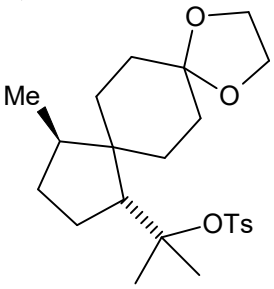
(A) 

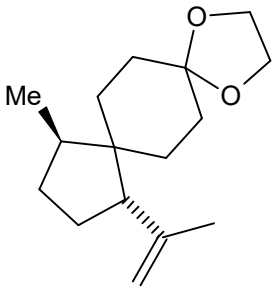
(B) 

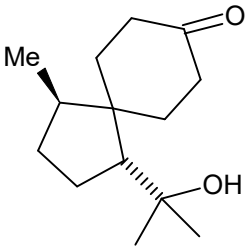
(C) 

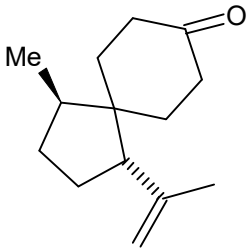
(D) 

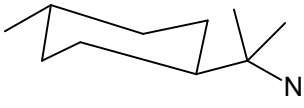
42.  $\xrightarrow[acetone]{TsOH}$ The major product

(A) 

(B) 

(C) 

(D) 

43.  $\xrightarrow[(ii)H_2O/\Delta]{(i)\ CH_3I(excess)}$ P

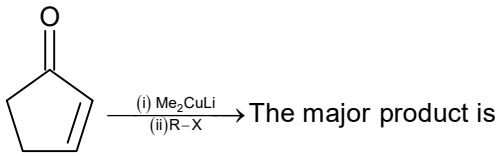
The correct statement about product P is

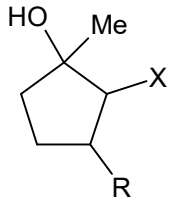
(A) The product is formed in according to Satzeff rule

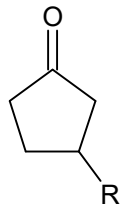
(B) The mechanism followed is E1

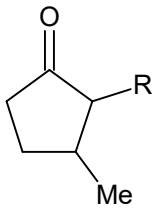
(C) The product has 1 degree of unsaturation and —OH group

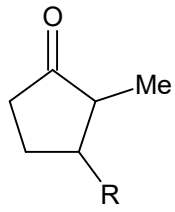
(D) The mechanism followed is E2

44. 

(A) 

(B) 

(C) 

(D) 

45. Sodium nitroprusside is used to confirm the presence of which element is organic compound?
(A) Nitrogen
(B) Nitrogen and sulphur
(C) Sulphur
(D) Halogen

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

46. The total number of cyclic isomer (position and optical) of a compound having molecule formula $C_3H_2Br_2$ is

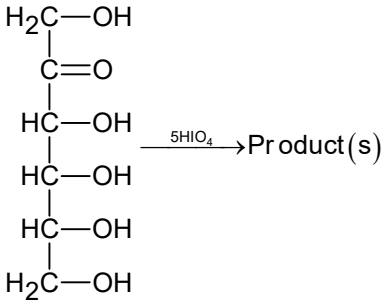
47. If the following compound is treated with Pd/C in excess of H_2 gas, how many stereoisomerism of the product will be obtained?

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

48. The isoelectric point for the given amino acid is:

$$(pk_a = 8.90)\ H_2N{-}\underset{\displaystyle (H_2C)_4{-}NH_2}{\overset{\displaystyle |}{CH}}{-}COOH\ (pk_a = 2.20)$$
$$(pk_a = 10.28)$$

49. 

The ratio of number of moles of CO_2 produced by HCOOH produced in the reaction is.

50. Considering the following polymers, if X is the number of addition polymer and Y is the number of condensation polymer, then calculate $\frac{X}{Y}$?

Dacron, Nylon 6, 6, Nylon-6, Teflon, Polyacrylonitrile

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

51. For how many integer values of k do the circle $x^2 + y^2 = k^2$ and hyperbola $xy = k$ not intersect each other
(A) 0
(B) 1
(C) 2
(D) 8

52. If F_1 and F_2 are the feet of the perpendiculars from the foci S_1 and S_2 of the ellipse $\frac{x^2}{36}+\frac{y^2}{25}=1$ on the tangent at any point P on the ellipse, then the minimum value of $S_1F_1 + S_2F_2$ is
(A) 18
(B) 10
(C) 30
(D) 25

53. From any point in the plane normals are drawn to the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$, if the eccentric angles of co-normals are $\alpha, \beta, \gamma, \delta$, then
(A) $\alpha+\beta+\gamma+\delta=\frac{\pi}{2}$
(B) $\alpha+\beta+\gamma+\delta=2\pi$
(C) $\alpha+\beta+\gamma+\delta=3\pi$
(D) $\alpha+\beta+\gamma+\delta=4\pi$

54. The vertices of a triangle are A(–1, –7), B(5, 1) and C(2, –3). If the internal angle bisector of $\angle B$ meets the side AC in D, then the length BD is
(A) $\frac{19}{3}$
(B) $\frac{20}{3}$
(C) $\frac{22}{3}$
(D) $\frac{23}{3}$

55. PQ is a variable line sliding between the coordinate axis in such a way that P lies on the x-axis and Q lies on the y-axis, if R is a variable point on PQ such that PR = 4, QR = 3 and PQ = 7, then the equation of the locus of R is
(A) $\frac{x^2}{3^2}+\frac{y^2}{4^2}=1$

(B) $\frac{x^2}{7^2}+\frac{y^2}{4^2}=1$

(C) $x^2+\frac{y^2}{3^2}=1$

(D) none of these

56. If origin is shifted such that the equation $x^2 + 7xy - 2y^2 + 17x - 26y - 60 = 0$ has no x and y terms, the coordinate of the new position is
(A) (–2, –3)
(B) (–2, 3)
(C) (2, –3)
(D) (3, 2)

57. An ellipse has the points (1, –1) and (2, –1) as its foci and x + y = 5 is one of its tangents, then the point of contact is
(A) $\left(\frac{11}{9},\frac{17}{9}\right)$
(B) $\left(\frac{17}{9},\frac{11}{9}\right)$
(C) $\left(\frac{11}{9},\frac{34}{9}\right)$
(D) $\left(\frac{34}{9},\frac{11}{9}\right)$

58. Let N be the foot of the perpendicular drawn from the focus S of any parabola to the tangent at any point P. If O be the vertex of the parabola, then
(A) OS, SN, SP are in AP
(B) OS, SN, SP are in GP
(C) OS, SN, SP are in HP
(D) none of these

59. Find the locus of focus of the family of parabolas having directrix of slope 2 and touching the line x = 1 and y = 3
(A) x + 2y = 5
(B) 2x + y = 5
(C) x + y = 5
(D) 2x – y – 5 = 0

60. Two tangents to a hyperbola $\frac{x^2}{2^2}-\frac{y^2}{3^2}=1$ meets its axes in concyclic points, then the locus of their point of intersection is
(A) $x^2 + y^2 = 13$
(B) $x^2 - y^2 = 4$
(C) $x^2 - y^2 = 13$
(D) $x^2 - y^2 = 5$

61. If sum of eccentric angle of two point on the hyperbola is $(2n + 1)\pi$, then chord passes through those two points always passes through which of the following points
(A) focus
(B) centre
(C) one of the ends of transverse axis
(D) one of the ends of conjugate axis

62. If I be the incentre of ΔABC, then $IA \cdot IB \cdot IC$ is, where R and r is circumradius and inradius
(A) $4Rr^2$
(B) $4R^2r$
(C) $4Rr$
(D) none of these

63. In a triangle ABC, a = 6, b = 3 and $\cos(A - B) = \frac{4}{5}$, then the angle C is
(A) $\frac{\pi}{4}$
(B) $\frac{\pi}{2}$
(C) $\frac{3\pi}{4}$
(D) $\frac{\pi}{6}$

64. The smallest and largest values of $\tan^{-1}\left(\frac{1-x}{1+x}\right)$, $0 \le x \le 1$ is
(A) $0, \pi$
(B) $0, \frac{\pi}{4}$
(C) $-\frac{\pi}{4}, \frac{\pi}{4}$
(D) $\frac{\pi}{4}, \frac{\pi}{2}$

65. $\sin^{-1}(3x - 4x^3)$ is
(A) $3\sin^{-1} x$, if $\frac{1}{2} \le x \le 1$
(B) $-\pi - 3\sin^{-1} x$, if $-\frac{1}{2} \le x \le \frac{1}{2}$
(C) $\pi - 3\sin^{-1} x$, $\frac{1}{2} < x \le 1$
(D) none of these

66. If $4 \sin x + 5 \cos ax = 9$ has at least one solution, then the possible values of a is, $(m, n \in z)$
(A) $a = 2m\pi$
(B) $a = \frac{4m\pi}{4n+1}$
(C) $a = (4n+1)\frac{\pi}{2}$
(D) $a = (2n+1)\frac{\pi}{2}$

67. Which of the following results is valid?
(A) $(1 + x)^n > (1 + nx)$ for all natural number n
(B) $(1 + x)^n \ge (1 + nx)$ for all natural number n
(C) $(1 + x)^n \le (1 + nx)$ for all natural number n
(D) $(1 + x)^n < (1 + nx)$ for all natural number n

68. The angle of elevation of the top of a vertical pole when observed from each vertex of a regular hexagon is $\frac{\pi}{3}$, if the area of a the circle circumscribing the hexagon be 8π m^2, then the area of the hexagon is

(A) $3\pi\sqrt{3}\,m^2$

(B) $8\sqrt{3}\,m^2$

(C) $6\sqrt{3}\,m^2$

(D) $12\sqrt{3}\,m^2$

69. Let ΔABC have side length AB = 10, BC = 12 and AC = 14, point x lies in the interior of $\overline{BC}$ and point I_1 and I_1 are the in-centres of ΔABX and ΔACX respectively. Then the minimum possible area of ΔAI_1I_2 as x varies alone BC

(A) 12

(B) 14

(C) 10

(D) 8

70. In ΔABC, AB = AC = 10 and BC = 12 and D lies on AB and E lies on AC such that AD = DE = EC, then AD can be

(A) $\frac{7}{25}$

(B) $\frac{250}{39}$

(C) $\frac{39}{250}$

(D) $\frac{25}{7}$

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

71. The tangent at P(2, 5) cuts the asymptotes of hyperbola with eccentricity $\sqrt{2}$ and centre C at (–1, 1) at Q and R, then the radius of circumcircle of ΔCQR is

72. The minimum value of $\frac{x^2\sin^2 x+9}{x\sin x}$, $\forall\ x\in\left(0,\frac{\pi}{2}\right)$

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

73. Non degenerate ΔABC has integer side lengths, BD is an angle bisector, AD = 3 and DC = 8. What is the smallest possible value of the perimeter?

74. In ΔABC, CD is the bisector of angle C, with D on AB, if $\cos\frac{C}{2}=\frac{1}{3}$ and CD = 6, then Harmonic mean of AC and BC is

75. The chord through (2, 1) to the circle $x^2 - 2x + y^2 - 2y + 1 = 0$ are bisected at the point $\left(\alpha, \frac{1}{2}\right)$, then α is